

ROTAX 912 iS / 912 iS SPORT LINE MAINTENANCE MANUAL (MML)

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****Authoritative OEM Specification — BRP-Rotax GmbH & Co KG / DRDO Certified****

****ATA 72-00 / ATA 73-10 / ATA 74-20 / ATA 76-00 / ATA 78-10 / ATA 79-00 / ATA 24-00****

CHAPTER 00-00: GENERAL SPECIFICATIONS & ENVELOPE LIMITS

1.1 Engine Technical Data

* ****Configuration:**** 4-cylinder, 4-stroke horizontally opposed boxer engine with liquid-cooled cylinder heads and ram-air cooled cylinders.

* ****Displacement:**** 1352 cm³ (82.5 cu in).

* ****Bore x Stroke:**** 84.0 mm x 61.0 mm.

* ****Compression Ratio:**** 10.8 : 1.

* ****Maximum Take-Off Power:**** 73.5 kW (100 HP) at 5,800 RPM (max 5 minutes).

* ****Maximum Continuous Power:**** 69.0 kW (92.5 HP) at 5,500 RPM.

* ****Idle Speed:**** 1,800 +/- 100 RPM.

* ****Propeller Reduction Ratio:**** 1 : 2.4286 (integrated dog-clutch overload mechanism).

1.2 Operational Limits

* ****Engine RPM:****

* Maximum Redline: 5,800 RPM.

* Maximum Continuous: 5,500 RPM.

* Cruise Range: 4,800 - 5,200 RPM.

* Economy Loiter: 4,300 - 4,600 RPM.

* ****Cylinder Head Temperature (CHT):****

* Nominal Range: 90 deg C - 115 deg C.

* Maximum Continuous: 135 deg C (275 deg F).

* Critical Alarm Threshold: > 140 deg C (Immediate descent / throttle reduction required).

* ****Exhaust Gas Temperature (EGT):****

* Nominal Cruise: 780 deg C - 820 deg C.

* Maximum Continuous: 880 deg C (1,616 deg F).

* Maximum Short Term (Take-Off): 950 deg C.

* Maximum Allowable Cylinder-to-Cylinder Disparity (Delta EGT): 65 deg C.

* ****Lubrication System (Oil Pressure & Temperature):****

* Nominal Oil Pressure: 2.0 - 5.0 bar (29 - 73 psi).

* Minimum Oil Pressure at Idle: 0.8 bar (12 psi).

* Maximum Cold Start Oil Pressure: 7.0 bar (102 psi).

* Nominal Oil Temperature: 90 deg C - 110 deg C.

* Minimum Oil Temperature for Take-Off: 50 deg C (122 deg F).

- * Maximum Oil Temperature: 130 deg C (266 deg F).
- * **Fuel Injection System:**
- * Fuel Rail Pressure: 3.0 +/- 0.2 bar (43.5 psi).
- * Fuel Flow at Cruise (5,000 RPM, Ladakh FL200): 14.5 - 17.5 L/h.
- * Fuel Flow at Economy Loiter: 11.0 - 13.5 L/h.
- * **Electrical System:**
- * Nominal DC Bus Voltage: 14.1 +/- 0.3 V.
- * Warning Voltage Low: < 12.8 V (Battery assisting bus).
- * Critical Voltage Low: < 12.0 V (Immediate load shedding mandatory).

CHAPTER 72-00: ENGINE CORE & AIRFLOW COOLING BAFFLES

2.1 Cylinder Head Cooling System

The Rotax 912 iS utilizes a hybrid cooling system: ram-air passes through specialized cowling baffles over the cylinder cooling fins, while an internal closed-loop water/glycol jacket cools the combustion chambers and valve seats.

2.2 Baffle Seal Maintenance & Inspection Procedures

1. **Inspection Interval:** Every 50 flight hours or after severe turbulence / temperature transitions.
2. **Seal Integrity:** Inspect the flexible elastomeric cooling baffle seals between the engine air plenum shroud and Cylinder #2 / Cylinder #4 cylinder heads.
3. **Failure Mode (DRDO Fault 1):** If the baffle seal becomes dislodged or deformed, ram airflow separates from the cylinder fin boundary layer, resulting in rapid CHT rise on Cylinder #2 (> 138 deg C).
4. **Corrective Action:**
 - * Step 1: Confirm CHT delta between Cyl #2 and Cyl #1 exceeds 25 deg C.
 - * Step 2: Enrich fuel trim +12% via GCS to lower combustion chamber temperature.
 - * Step 3: Reduce throttle to 4,600 RPM.
 - * Step 4: Inspect baffle clip tension and replace torn silicone seals with OEM Part No. 965-021.

CHAPTER 73-10: HIGH PRESSURE FUEL INJECTION (FADEC)

3.1 Dual Injector Architecture

Each cylinder runner is equipped with two electromagnetic high-pressure fuel injectors (Injector A and Injector B) controlled independently by FADEC Lane A and Lane B.

3.2 Fuel Injector Clog Diagnostics (DRDO Fault 2)

1. **Symptom Signature:** Total fuel flow rate drops by 15-25%, while Cylinder #1 EGT spikes above 880 deg C due to extreme lean-burn combustion.
2. **Kinematic Effect:** Unequal torque stroke distribution induces cyclic crankshaft deceleration, dropping engine RPM by 60-120 RPM and elevating reduction gearbox vibration (VIB > 1.2 mm/s).
3. **Emergency SOP:**

- * Force FADEC arbitration to Lane B auxiliary injection schedule.
- * Engage auxiliary electric fuel boost pump to ensure minimum 3.0 bar rail pressure.
- * If EGT remains divergent, land at nearest recovery airfield.
- 4. ****Maintenance Action:**** Disassemble fuel rail, remove Injector #1 (Part No. 874-320), and perform ultrasonic solvent backflush and flow rate calibration (nominal 185 cc/min at 3.0 bar).

CHAPTER 74-20: DUAL IGNITION & SPARK PLUG SYSTEM

4.1 System Overview

Dual capacitive discharge ignition system with two spark plugs per cylinder (top and bottom). Driven by dual internal stator generators and FADEC ignition timing maps.

4.2 Ignition Misfire Diagnostics (DRDO Fault 3)

1. ****Symptom Signature:**** Unburnt fuel expelled into exhaust runner causes sharp EGT drop on affected cylinder (-90 to -135 deg C), accompanied by high-variance RPM flutter (+/- 150 RPM) and intense gearbox torsional shock vibration (VIB > 2.0 mm/s).
2. ****Root Cause:**** Primary ignition coil insulation breakdown or spark plug lead carbon track breakdown on Lane A circuit.
3. ****Emergency Action:**** Switch ignition arbitration from Lane A to Lane B via GCS control switch.
4. ****Post-Flight Order:**** Perform insulation resistance check on ignition lead (nominal 4.5 - 5.5 kOhm). Replace spark plug pack with OEM Part No. 897-257.

CHAPTER 79-00: DRY-SUMP LUBRICATION SYSTEM

5.1 Hydraulic Circuit

Dry-sump system with integrated mechanical pressure pump, scavenge return lines, external oil reservoir (3.0 L capacity), and thermostat-controlled oil cooler.

5.2 Oil Pressure Loss & Bearing Failure (DRDO Fault 4)

1. ****Symptom Signature:**** Oil pressure collapses below 2.0 bar (typically falling to 1.6 - 1.8 bar). Hydrodynamic oil wedge breakdown causes severe bearing friction, resulting in rapid oil temperature climb (> 120 deg C), conduction heat rise on all 4 cylinder heads (+10 deg C), and parasitic RPM drag.
2. ****Emergency Action:**** Reduce throttle to 4,200 RPM immediately to minimize bearing shear stress. Initiate immediate precautionary landing / recovery vector.
3. ****Maintenance Action:**** Remove and inspect suction screen, magnetic drain plug (Part No. 941-895), and oil filter for bronze/steel metallic shavings.